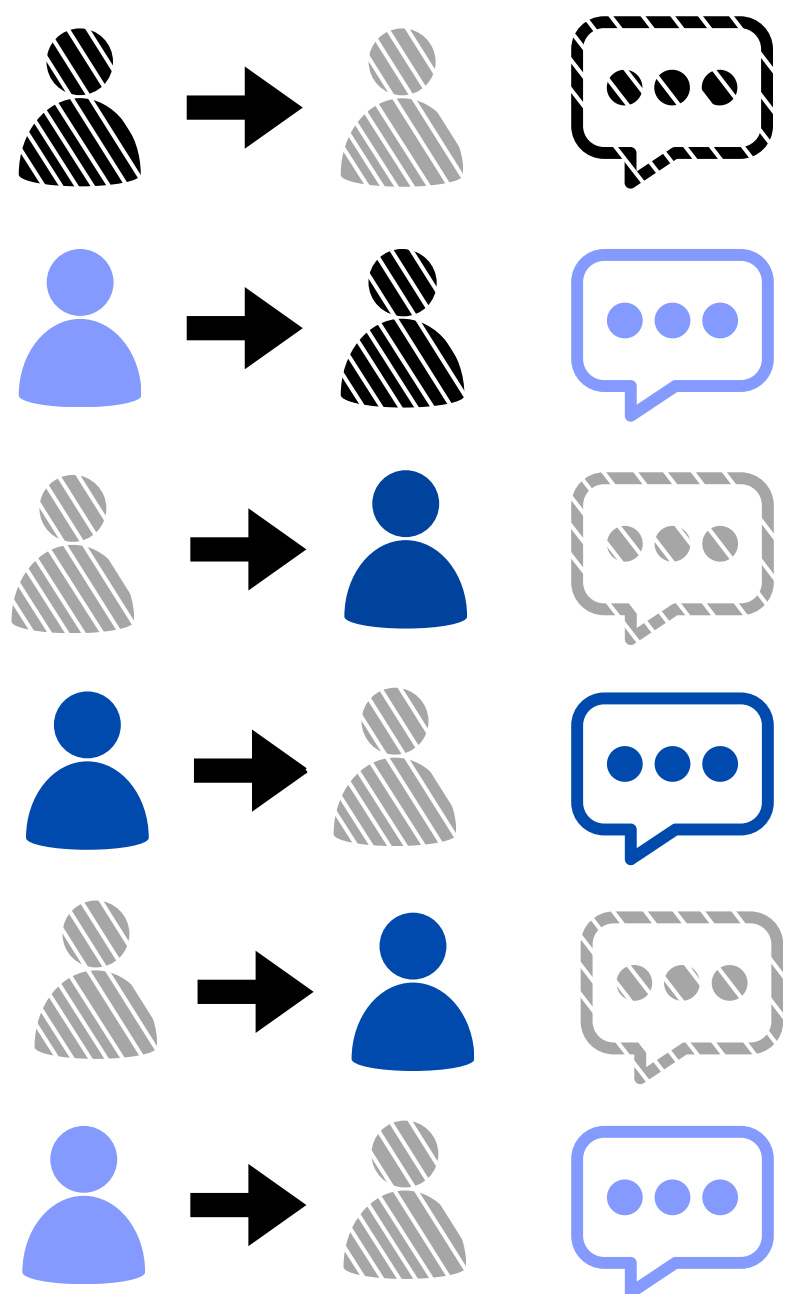


# Do LLMs suffer from Multi-Party Hangover?

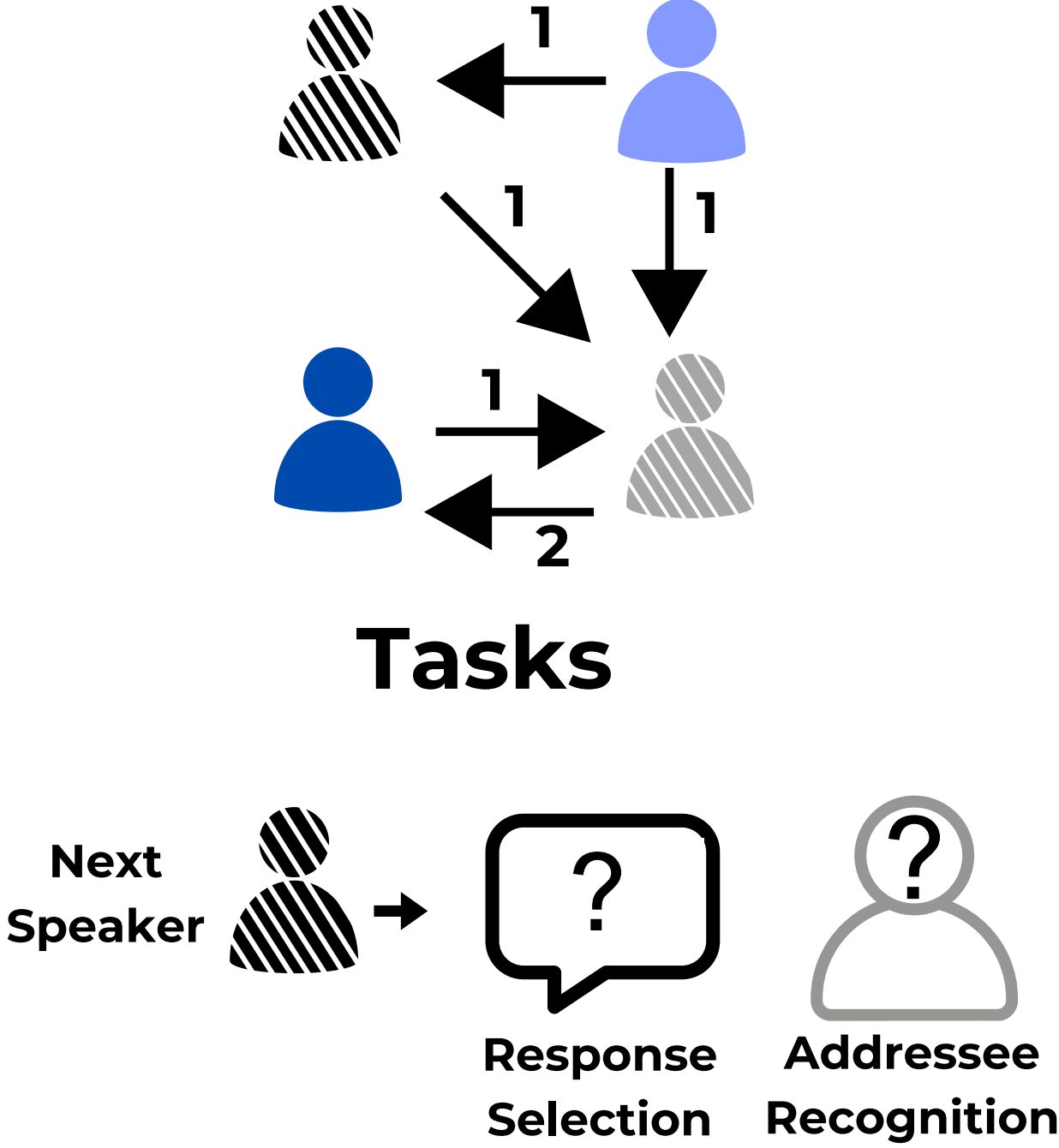
## A Diagnostic Approach to Addressee Recognition and Response Selection in Conversations

Nicolò Penzo, Maryam Sajedinia, Bruno Lepri, Sara Tonelli, Marco Guerini

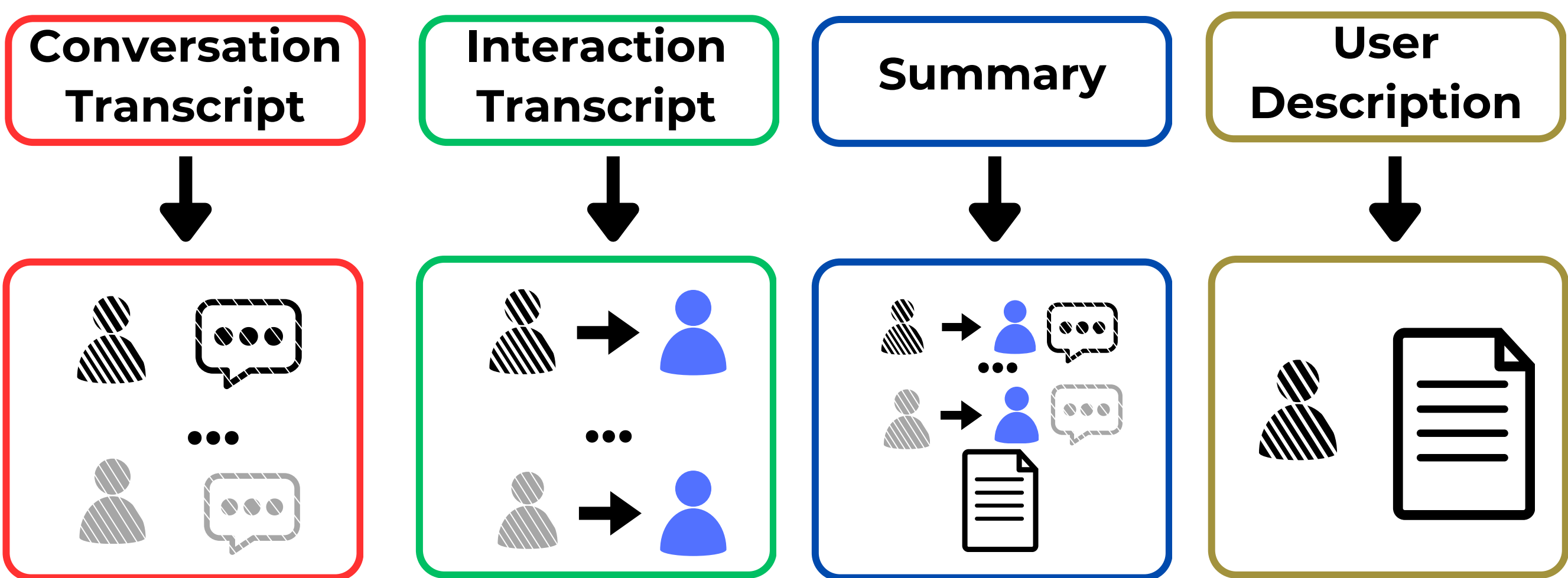
### Multi-Party Conversation



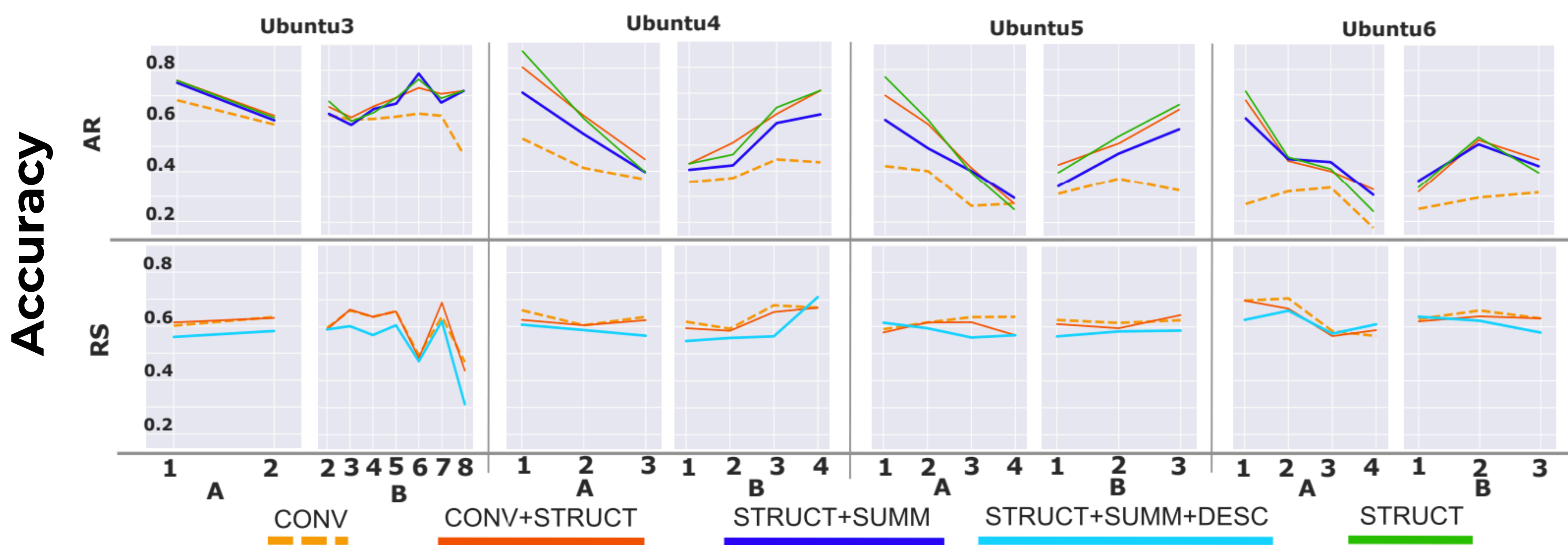
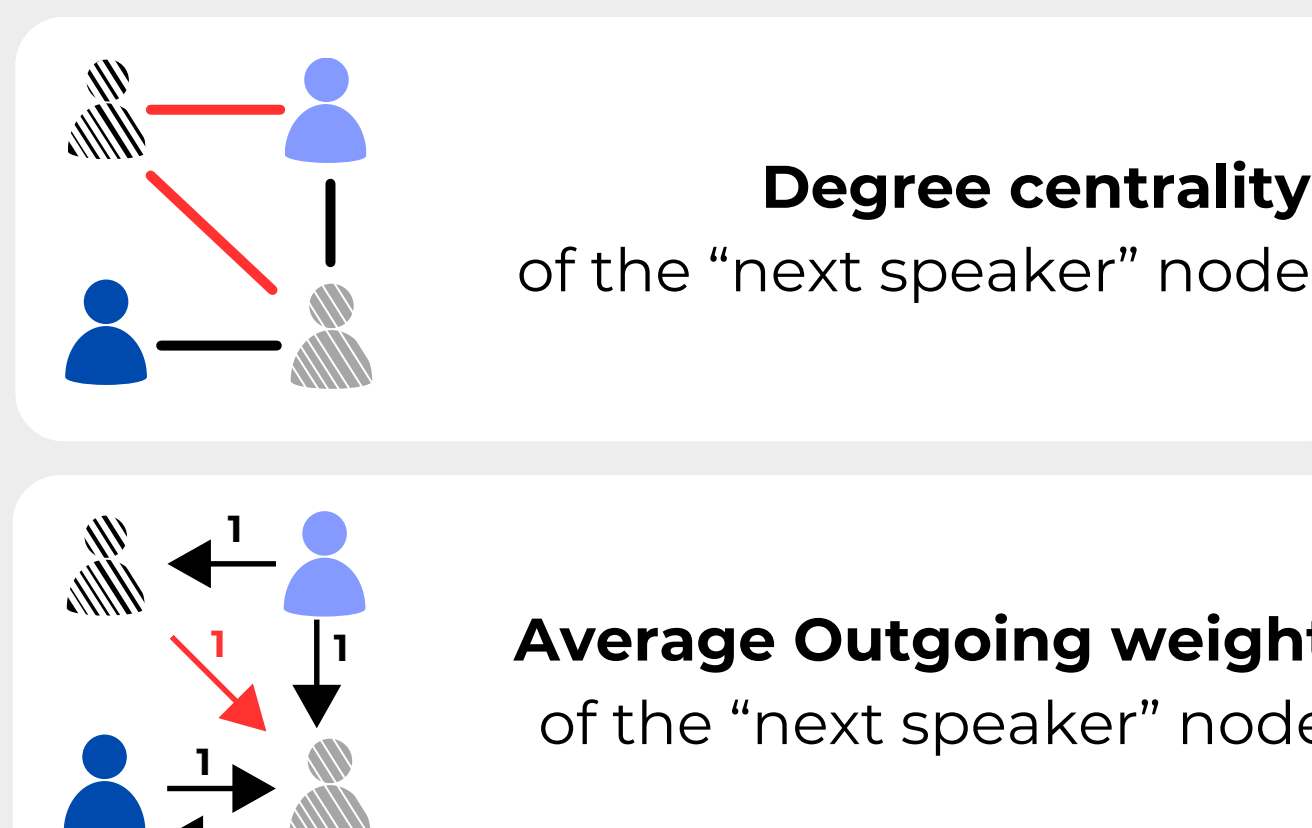
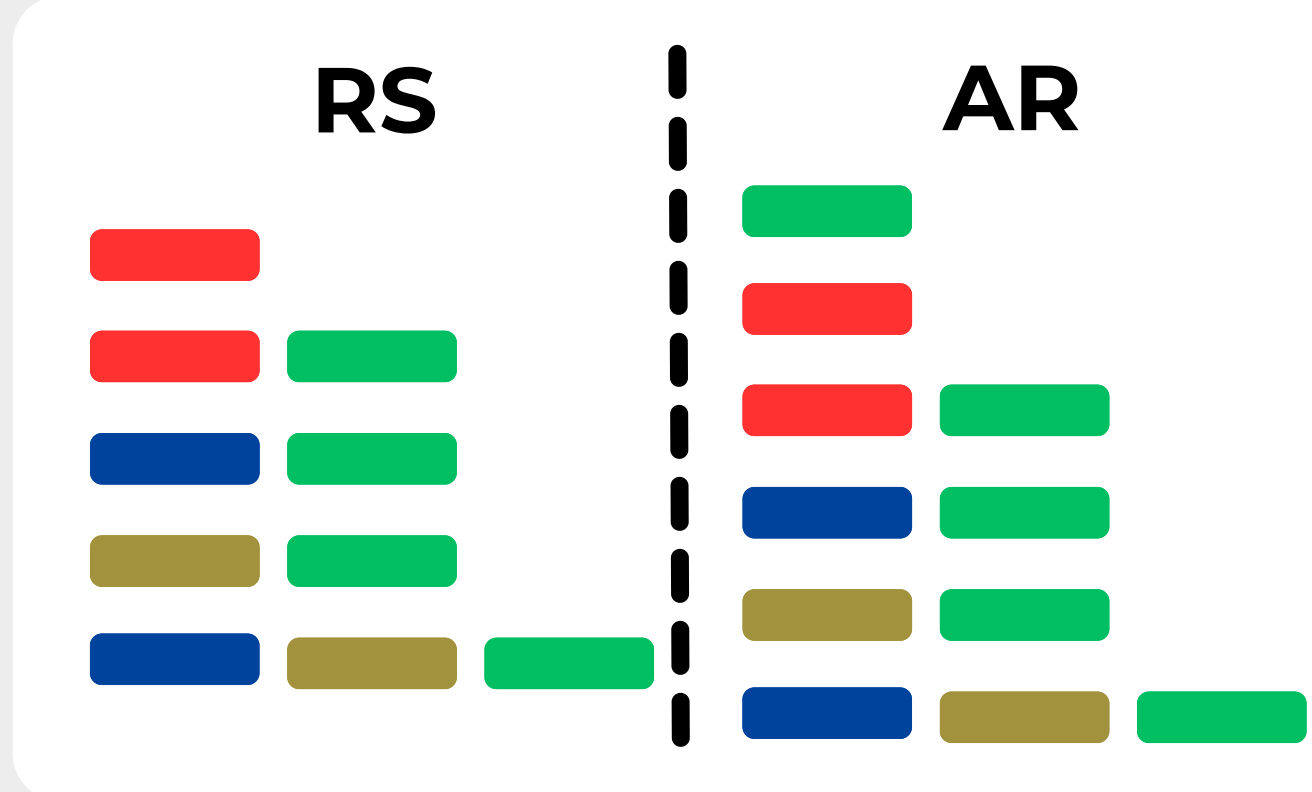
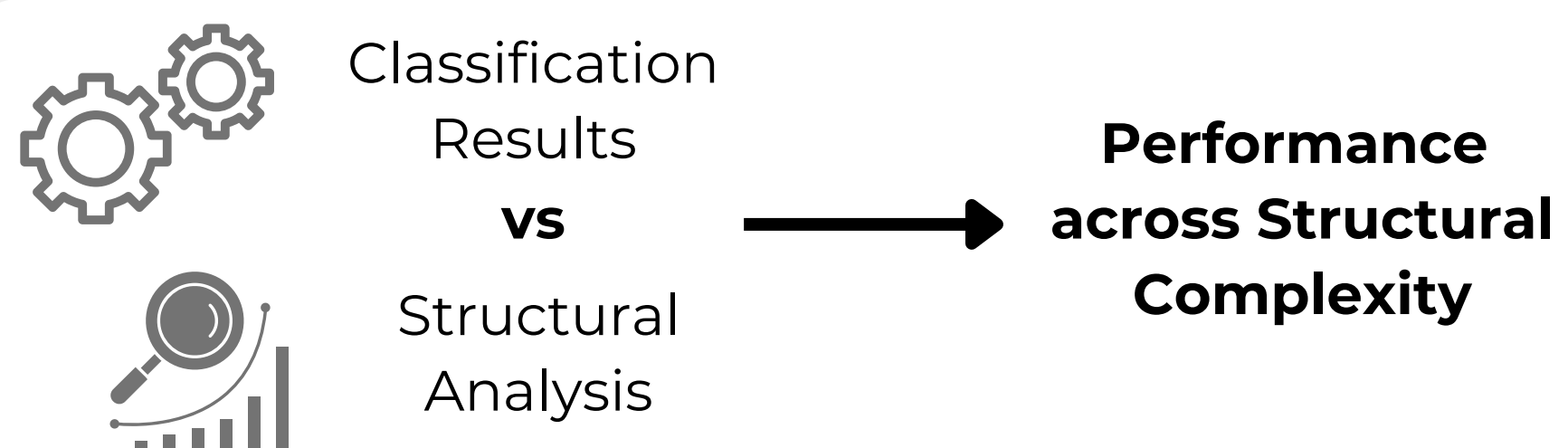
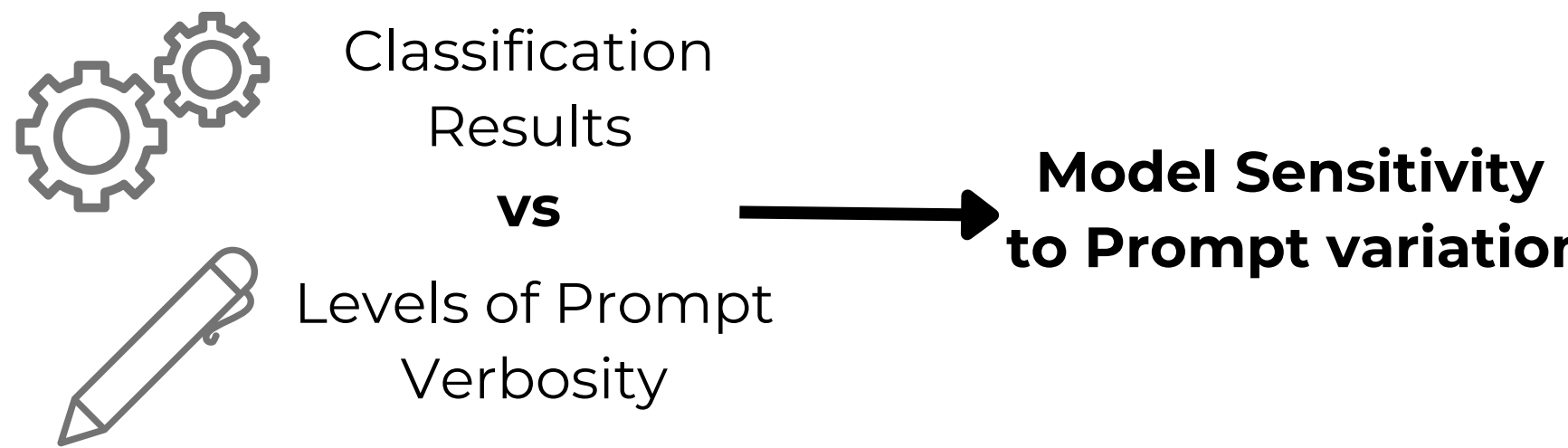
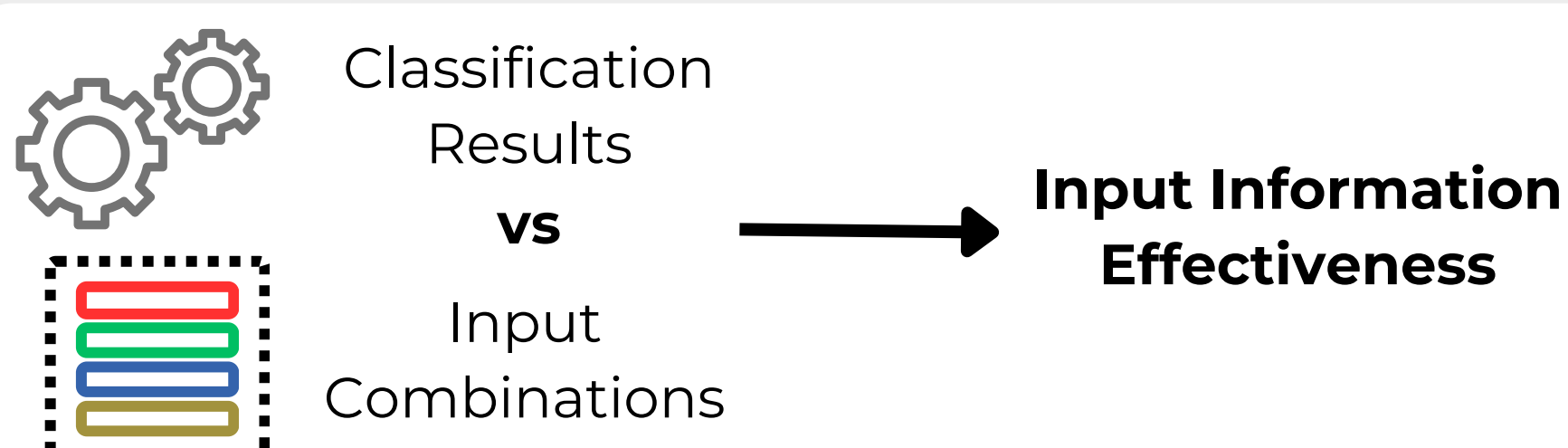
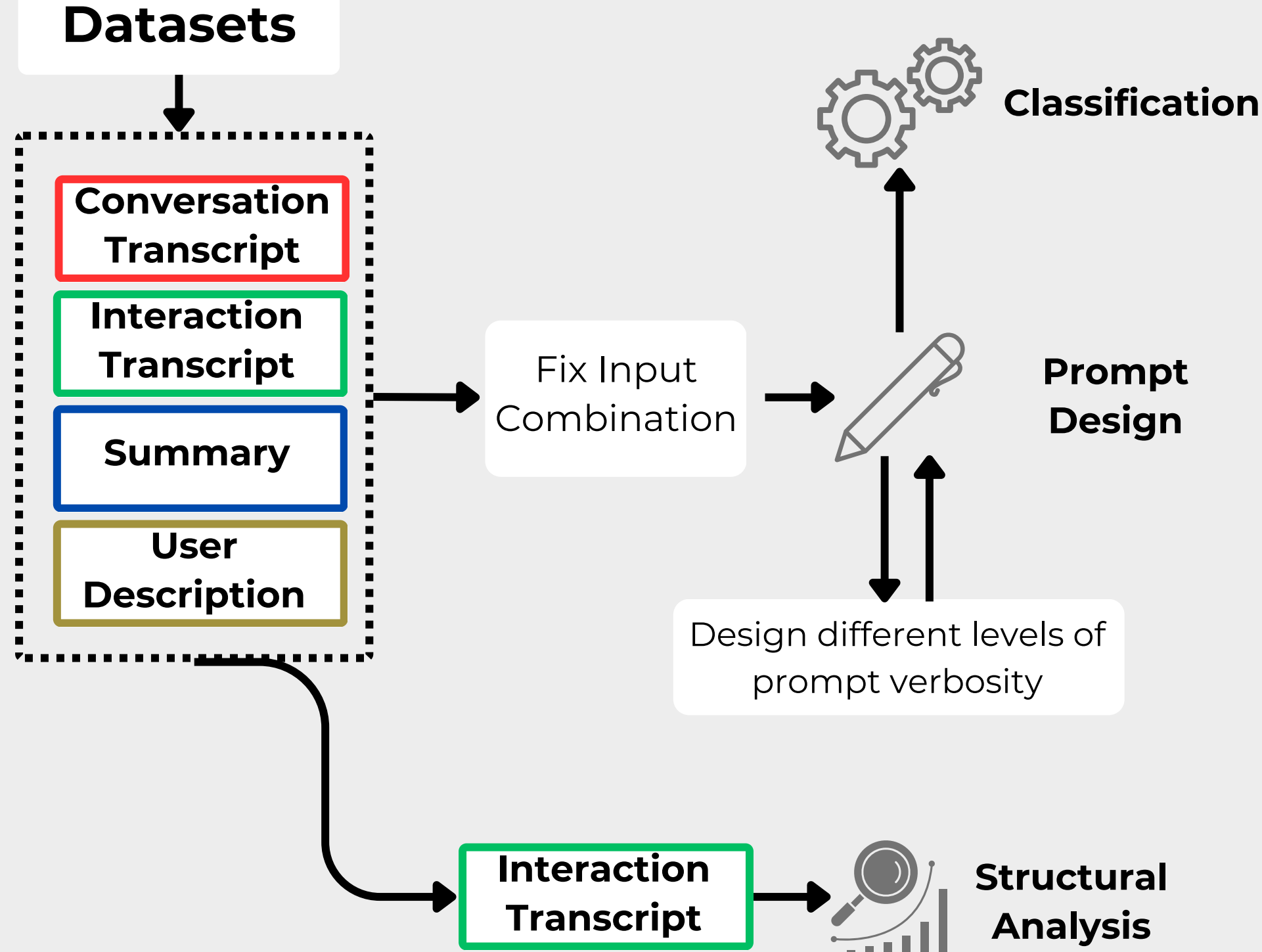
### Interaction Graph



### Input Information



### Diagnostic Datasets



A = Degree centrality  
B = Average Outgoing weight

### FINDINGS

- **Original** information is **more effective** both in Response Selection (RS) and Addressee Recognition (AR). **Interaction transcripts** are **crucial** for the **structural task** (AR), but **not** for the **linguistic task** (RS).
- The **structural task** (AR) shows **greater sensitivity** to prompts verbosity **compared** to the **linguistic task** (RS).
- In the **structural task** (AR), performance **differences** across **input combinations** (always including Interaction Transcript) are due to **improved outcomes** on items of **low structural complexity**.

EMNLP  
2024

### Reference:

Nicolò Penzo, Maryam Sajedinia, Bruno Lepri, Sara Tonelli, and Marco Guerini. 2024. *Do LLMs suffer from Multi-Party Hangover? A Diagnostic Approach to Addressee Recognition and Response Selection in Conversations*. In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing. Association for Computational Linguistics.

### REPO



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